Variation

Changing examples to deepen understanding

Use variation to help students notice the structure of an idea rather than copy surface features. Decide what changes, what stays the same and why that comparison matters.

How to do it well

1 Choose the relationship. Identify the feature, rule or structure students should notice across examples.

2 Establish a baseline. Solve or explain one accessible example so the comparison has a clear starting point.

3 Change a consequential feature. Keep other demands stable and ask students to predict the effect.

4 Compare the outcomes. Discuss what changed and what remained valid, using the underlying principle rather than appearance.

5 Test the generalisation. Present a less familiar case and ask students to justify whether the same relationship still applies.

What this sounds like

“Only the direction of the arrow has changed. Predict what that changes in the explanation, and identify what remains true.”

Classroom example. A Science teacher changes one condition in a familiar investigation and asks pupils to predict the consequence before calculating or explaining. Pupils compare the new outcome with the baseline and identify the relationship that remains true. A further example tests the generalisation, so variation directs attention to structure rather than mere novelty.

Top tips that make the technique work

Change deliberately. A sequence is not purposeful variation simply because every question looks different.

Include an unchanged relationship. Students need something stable against which to notice the new feature.

Ask for a prediction. Before calculation or completion, ask what they expect to change and why.

Use near-misses. A small change that makes a case invalid can clarify a concept boundary.

Compare after solving. Ask students to explain the effect of the change rather than moving straight to another item.

Do and avoid

Do	Avoid
Select changes that reveal the target relationship.	Changing several unfamiliar features at once.
Ask students to describe the invariant.	Assuming a varied worksheet automatically produces conceptual understanding.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation. Some links support part of this guide rather than the whole technique.

Barton - Purposeful Practice | DistillED - Non-Examples